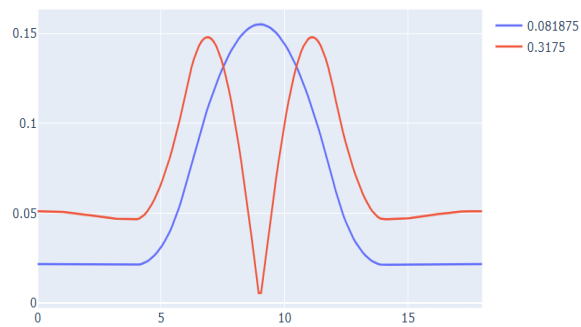
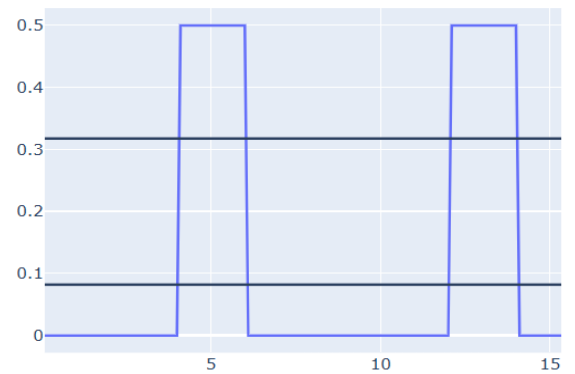


EXP-6: Demonstration of Quantum Tunneling

Wave-functions from tight-binding

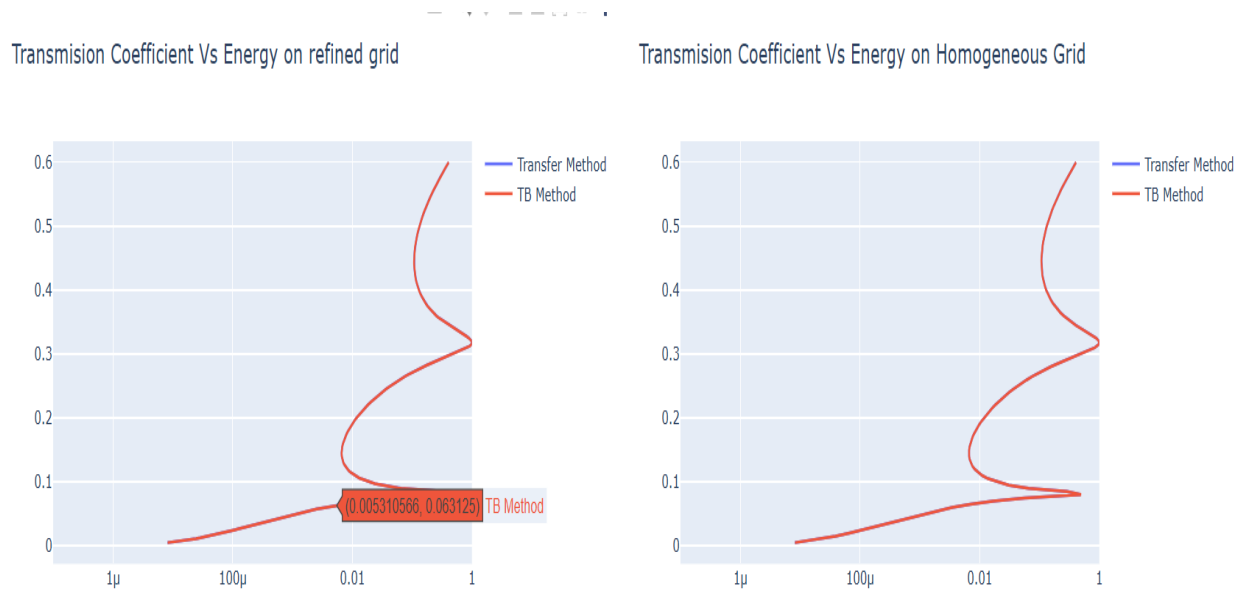


Potential Energy vs Distance



1. This graph shows the **wave functions** obtained using the **Tight-Binding (TB) method**.
2. The blue and red curves represent the **electron probability distribution** at different energy states.
3. The peaks indicate regions where electrons are **most likely to be found**.
4. The variation in the wave functions reflects the **quantum confinement effect** inside the structure.
5. This analysis helps in designing **quantum wells, nanowires, quantum dots, and semiconductor devices**.

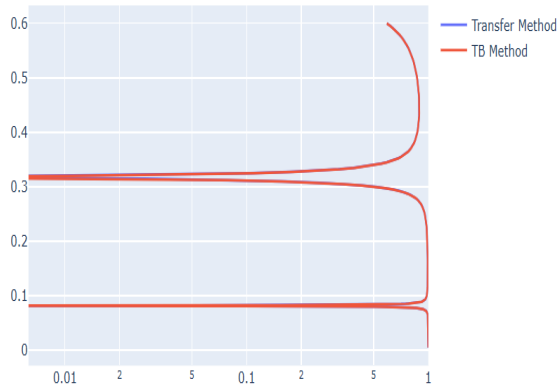
Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid



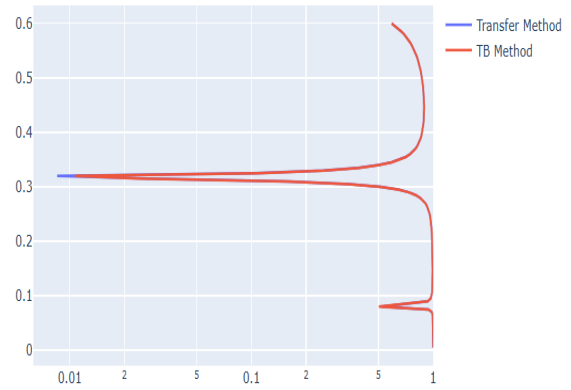
1. This figure compares the **Transmission Coefficient vs Energy** using the **Transfer Matrix Method** and the **Tight-Binding (TB) Method** on both **refined** and **homogeneous** grids.
2. The blue and red curves show the transmission behavior predicted by the two numerical methods.
3. The close overlap of the curves indicates that both methods produce **consistent and accurate results**.
4. The variation in the transmission coefficient with energy reveals the **allowed and restricted electron transport regions**.
5. This analysis is useful for designing **quantum devices, semiconductor nanostructures, resonant tunneling devices, and nanoelectronic systems**.

Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid

Reflection Coefficient Vs Energy on refined grid



Reflection Coefficient Vs Energy on Homogeneous Grid



1. This figure compares the **Reflection Coefficient vs Energy** using the **Transfer Matrix Method** and the **Tight-Binding (TB) Method** on both **refined** and **homogeneous** grids.
2. The blue and red curves represent the reflection behavior of electrons at different energy levels.
3. The close agreement between the two methods confirms the **accuracy and reliability** of the simulation results.
4. The variation in the reflection coefficient identifies the **energy regions where electrons are strongly or weakly reflected** by the potential barriers.
5. This analysis is important for designing **quantum transport devices, resonant tunneling structures, semiconductor nanodevices, and nanoelectronic circuits**.